Team 28

Hello! In this report, we will present our final project – Book Management System.

Login/Registration

The MyGUI class is a Java program that creates a simple login and registration form using Swing. Its purpose is to allow users to either log in with existing credentials or register new accounts. The program stores user data in a CSV file and provides basic functionality for user authentication and registration.

Code Snippets:

```
// Registration heading
JLabel registrationLabel = new JLabel(text:"Registration");
registrationLabel.setFont(new Font(name:"Arial", Font.BOLD, size:28));
registrationLabel.setForeground(Color.BLACK); // Change color to black
contentPane.add(registrationLabel, gbcCenter);

// Username label and field
JLabel usernameLabel = new JLabel(text:"Username:");
JTextField usernameField = new JTextField();
usernameField.setPreferredSize(new Dimension(width:200, height:30)); // Set preferred size
GridBagConstraints gbcUsernameLabel = new GridBagConstraints();
gbcUsernameLabel.anchor = GridBagConstraints.EAST;
contentPane.add(usernameLabel, gbcUsernameLabel);
contentPane.add(usernameField, gbcCenter);

// Password label and field
JLabel passwordLabel = new JLabel(text:"Password:");
JPasswordField passwordField = new JPasswordField();
passwordField.setPreferredSize(new Dimension(width:200, height:30)); // Set preferred size
GridBagConstraints gbcPasswordLabel = new GridBagConstraints();
gbcPasswordLabel.anchor = GridBagConstraints.EAST;
contentPane.add(passwordLabel, gbcPasswordLabel);
contentPane.add(passwordField, gbcCenter);
```

```java
// Register button
JButton registerButton = new JButton(text:"Register");
registerButton.setBackground(new Color(r:34, g:139, b:34)); // Dark green
registerButton.setForeground(Color.WHITE);
registerButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        String username = usernameField.getText();
        String password = new String(passwordField.getPassword());

        // Validate input fields
        if (username.isEmpty() || password.isEmpty()) {
            JOptionPane.showMessageDialog(MyGUI.this, message:"Please enter both username and password.", title:"Error", JOptionPane.ERROR_MESSAGE);
            return;
        }

        // Check if username already exists
        if (usersMap.containsKey(username)) {
            JOptionPane.showMessageDialog(MyGUI.this, message:"Username already exists. Please choose a different username.", title:"Error", JOptionPane.ERROR_MESSAGE);
        } else {
            // Check password strength
            if (!isPasswordStrong(password)) {
                JOptionPane.showMessageDialog(MyGUI.this,
                        message:"Password must be at least 8 characters long and contain at least one uppercase letter, one lowercase letter, one digit, and one special character.",
                        title:"Error", JOptionPane.ERROR_MESSAGE);
                return;
            }

            // Add new user to the map and update CSV file
            User newUser = new User(username, password, isAdmin:false);
            usersMap.put(username, newUser);
            saveUsersToCSV();
            JOptionPane.showMessageDialog(MyGUI.this, message:"Registration successful. You can now login.", title:"Success", JOptionPane.INFORMATION_MESSAGE);

            // Clear username and password fields after successful registration
            usernameField.setText(t:"");
            passwordField.setText(t:"");
        }
    }
});
```

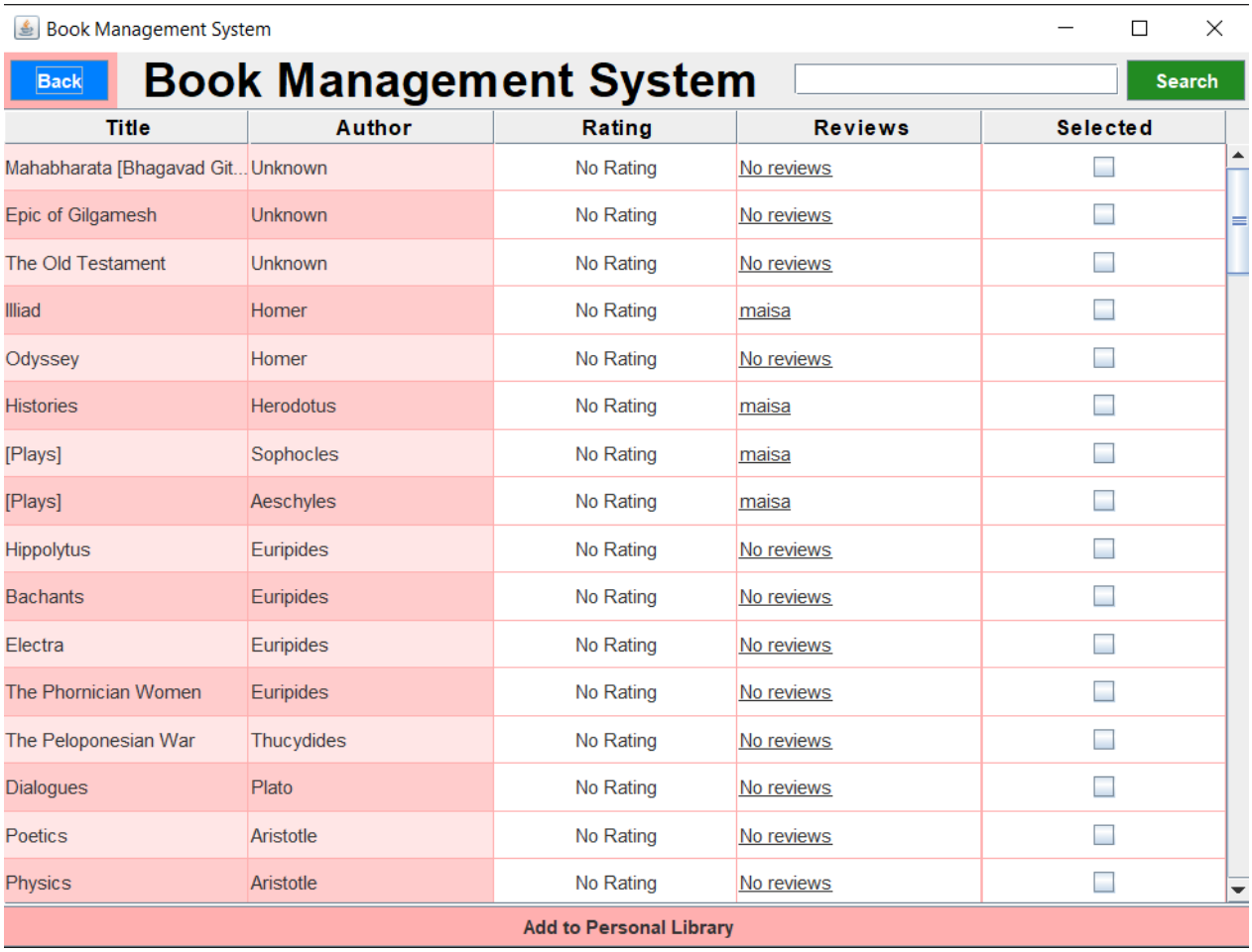
```java
// Login button
JButton loginButton = new JButton(text:"Login");
loginButton.setBackground(new Color(r:0, g:128, b:255)); // Dark blue
loginButton.setForeground(Color.WHITE);
loginButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        String username = usernameField.getText();
        String password = new String(passwordField.getPassword());

        // Validate login credentials
        if (usersMap.containsKey(username)) {
            User user = usersMap.get(username);
            if (user.verifyPassword(password)) {
                if (user.isAdmin()) {
                    // Open the ShowBooksAdmin page for admin user
                    new GeneralDatabaseOfAdmin().setVisible(b:true);
                } else {
                    // Open the ShowBooks page for non-admin user
                    new GeneralDatabaseGUI(username).setVisible(b:true);
                }
                // Close the login/registration page
                dispose();
            } else {
                JOptionPane.showMessageDialog(MyGUI.this, message:"Invalid username or password. Please try again.", title:"Error", JOptionPane.ERROR_MESSAGE);
            }
        } else {
            JOptionPane.showMessageDialog(MyGUI.this, message:"Invalid username or password. Please try again.", title:"Error", JOptionPane.ERROR_MESSAGE);
        }
    }
});
```

UI Snippet:

General Database

The code creates a user interface for managing books.

UI Snippet:



Title	Author	Rating	Reviews	Selected
Mahabharata [Bhagavad Git...	Unknown	No Rating	No reviews	☐
Epic of Gilgamesh	Unknown	No Rating	No reviews	☐
The Old Testament	Unknown	No Rating	No reviews	☐
Illiad	Homer	No Rating	maisa	☐
Odyssey	Homer	No Rating	No reviews	☐
Histories	Herodotus	No Rating	maisa	☐
[Plays]	Sophocles	No Rating	maisa	☐
[Plays]	Aeschyles	No Rating	maisa	☐
Hippolytus	Euripides	No Rating	No reviews	☐
Bachants	Euripides	No Rating	No reviews	☐
Electra	Euripides	No Rating	No reviews	☐
The Phornician Women	Euripides	No Rating	No reviews	☐
The Peloponesian War	Thucydides	No Rating	No reviews	☐
Dialogues	Plato	No Rating	No reviews	☐
Poetics	Aristotle	No Rating	No reviews	☐
Physics	Aristotle	No Rating	No reviews	☐

Search

It starts by setting up a search panel with a text field and a button for searching titles or authors. The table below displays book information such as title, author, rating, and reviews, with a checkbox column for selection.

Code Snippet:

```java
// Search panel with FlowLayout
    JPanel searchPanel = new JPanel(new FlowLayout(FlowLayout.RIGHT));

    // Search field
    searchField = new JTextField(20);
    searchField.setToolTipText("Enter title or author to search");
    searchPanel.add(searchField);

    // Search button
    JButton searchButton = new JButton("Search");
    searchButton.setBackground(new Color(34, 139, 34)); // Dark green
    searchButton.setForeground(Color.WHITE);
    searchButton.addActionListener(e -> {
        String query = searchField.getText().trim().toLowerCase();
        filterBooks(query);
    });

    searchPanel.add(searchButton);

    // Add search panel to NORTH of the frame
    headerPanel.add(searchPanel, BorderLayout.EAST);
```

Table

The program reads data from a CSV file to populate the table initially. It customizes the appearance of the table, including row colors and alignment of certain columns. Users can click on the "Reviews" column to view detailed reviews by selecting a specific user from a dropdown **menu.**

Code Snippet:

```java
// Initialize table model
tableModel = new DefaultTableModel() {
    @Override
    public Class<?> getColumnClass(int columnIndex) {
        if (columnIndex == 4) {
            return Boolean.class;
        } else {
            return String.class;
        }
    }

    @Override
    public boolean isCellEditable(int row, int column) {
        return column == 4;
    }
};
tableModel.addColumn("Title");
tableModel.addColumn("Author");
tableModel.addColumn("Rating");
tableModel.addColumn("Reviews");
tableModel.addColumn("Selected");

// Read data from CSV and populate the table
readDataFromCSV();

// Table setup
JTable bookTable = new JTable(tableModel);
bookTable.setRowHeight(30);
bookTable.setGridColor(Color.PINK);

// Customize column header appearance
JTableHeader tableHeader = bookTable.getTableHeader();
tableHeader.setFont(new Font("Arial", Font.BOLD, 14));
tableHeader.setForeground(Color.BLACK);

// Center-align "Rating" and "Reviews" columns
DefaultTableCellRenderer centerRenderer = new DefaultTableCellRenderer();
centerRenderer.setHorizontalAlignment(JLabel.CENTER);
bookTable.getColumnModel().getColumn(2).setCellRenderer(centerRenderer);
bookTable.getColumnModel().getColumn(3).setCellRenderer(centerRenderer);

// Customize row colors
bookTable.setDefaultRenderer(Object.class, new DefaultTableCellRenderer() {
    @Override
    public Component getTableCellRendererComponent(JTable table, Object value,
                                                   boolean isSelected, boolean hasFocus,
                                                   int row, int column) {
        Component c = super.getTableCellRendererComponent(table, value, isSelected, hasFocus, row, column);
        c.setBackground(row % 2 == 0 ? new Color(255, 230, 230) : new Color(255, 204, 204)); // Alternating shades of pink for rows
        return c;
    }
});

bookTable.getColumnModel().getColumn(3).setCellRenderer(new ReviewCellRenderer());
bookTable.addMouseListener(new MouseAdapter() {
    @Override
    public void mouseClicked(MouseEvent e) {
        int column = bookTable.getColumnModel().getColumnIndex("Reviews");
        int row = bookTable.rowAtPoint(e.getPoint());

        if (column == 3 && row != -1) { // Clicked on review column and valid row
            Rectangle cellRect = bookTable.getCellRect(row, column, false);
            if (cellRect.contains(e.getPoint())) { // Check if click is within the cell bounds
                String reviews = (String) bookTable.getValueAt(row, column);
                if (!reviews.equals("No reviews")) {
                    String[] users = reviews.split(", ");
                    String selectedUser = (String) JOptionPane.showInputDialog(
                            GeneralDatabaseGUI.this,
                            "Select a user to view details:",
                            "User Details",
                            JOptionPane.PLAIN_MESSAGE,
                            null,
                            users,
                            null);

                    if (selectedUser != null && !selectedUser.isEmpty()) {
                        // Get book details
                        String title = (String) bookTable.getValueAt(row, 0);
                        String author = (String) bookTable.getValueAt(row, 1);
                        String averageRating = (String) bookTable.getValueAt(row, 2);

                        // Get user's details for the selected book
                        String userRating = getUserRatingForBook(selectedUser, title);
                        String userReview = getUserReviewForBook(selectedUser, title);

                        // Display user's details in a separate window
                        showUserDetails(title, author, averageRating, selectedUser, userRating, userReview);
                    }
                }
            }
        }
    }
});
```

UI Snippet:

Title	Author	Rating	Reviews	Selected
Mahabharata [Bhagavad Git...	Unknown	No Rating	No reviews	☐
Epic of Gilgamesh	Unknown	No Rating	No reviews	☐
The Old Testament	Unknown	No Rating	No reviews	☐
Illiad	Homer	No Rating	maisa	☐
Odyssey	Homer	No Rating	No reviews	☐
Histories	Herodotus	No Rating	maisa	☐
[Plays]	Sophocles	No Rating	maisa	☐
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Electra	Euripides	No Rating	No reviews	☐
The Phornician Women	Euripides	No Rating	No reviews	☐
The Peloponesian War	Thucydides	No Rating	No reviews	☐
Dialogues	Plato	No Rating	No reviews	☐
Poetics	Aristotle	No Rating	No reviews	☐
Physics	Aristotle	No Rating	No reviews	☐

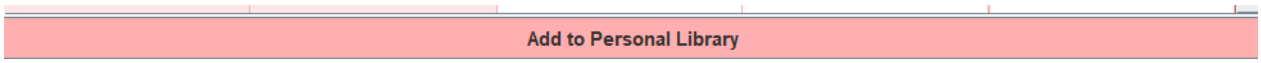
Add to library

At the bottom of the interface, there's a button to add selected books to the user's personal library. Finally, the GUI is set to be visible to the user.

Code Snippet:

```
        // Add "Add to Library" button
         JButton addToLibraryButton = new JButton("Add to Personal Library");
         addToLibraryButton.setBackground(Color.PINK);
         addToLibraryButton.addActionListener(e -> addToLibrary(username));
         add(addToLibraryButton, BorderLayout.SOUTH);
dispose();
FilterAndSortFunction.SortSelected(bookTable);
         setVisible(true);
```

UI Snippet:



Personal Database:

The provided Java code describes the implementation of a GUI application using Swing, specifically tailored for managing a personal library database. This application allows users to add, edit, and delete book entries, as well as track their reading progress. Here's a detailed analysis of the key components and functionalities:

Code Snippets:

```
setTitle(title:"Add to library");
setSize(width:800, height:400);
setLocationRelativeTo(c:null);

// Setting up the username label at the top of the window
JLabel usernameLabel = new JLabel("Username: " + username);
usernameLabel.setFont(new Font(name:"Arial", Font.BOLD, size:16));
usernameLabel.setHorizontalAlignment(SwingConstants.CENTER);
usernameLabel.setOpaque(isOpaque:true);
usernameLabel.setBackground(Color.PINK);
usernameLabel.setForeground(Color.BLACK);

// Panel for username label and back button
JPanel topPanel = new JPanel(new BorderLayout());
topPanel.setBackground(Color.PINK);
topPanel.add(usernameLabel, BorderLayout.CENTER);

// Back button to return to the previous screen
JButton backButton = new JButton(text:"Back");
backButton.addActionListener(e -> {
    dispose();
    GeneralDatabaseGUI showBooks = new GeneralDatabaseGUI(username);
    showBooks.setVisible(b:true);
});
backButton.setBackground(new Color(r:0, g:128, b:255)); // Dark blue
backButton.setForeground(Color.WHITE);
topPanel.add(backButton, BorderLayout.WEST);
getContentPane().add(topPanel, BorderLayout.NORTH);
```

```java
    // Table setup with specified column names
    String[] columnNames = {"Title", "Author", "Rating", "Reviews", "Status", "Spend Time (minutes)", "Start Date", "End Date", "User Rating", "User Review", "Delete"};
    tableModel = new DefaultTableModel(null, columnNames) {
        @Override
        public boolean isCellEditable(int row, int column) {
            return column >= 4; // Make columns from "Status" onwards editable
        }
    };

    // Adding books to the table
    for (String[] book : selectedBooks) {
        Object[] rowData = {book[0], book[1], book[2], book[3], "Not Started", "", "", "", "Add rating", "Add review", "Delete"};
        tableModel.addRow(rowData);
    }
    for (String[] book : userBookHistory) {
        tableModel.addRow(book);
    }

    // Creating the table and setting its properties
    JTable addToLibraryTable = new JTable(tableModel);
    addToLibraryTable.setRowHeight(rowHeight:30);

    // Custom renderers to handle display of non-editable columns
    for (int i = 0; i < 4; i++) {
        addToLibraryTable.getColumnModel().getColumn(i).setCellRenderer(new NonEditableCellRenderer());
    }

    // Custom cell editors for various columns to handle specific data types
    addToLibraryTable.getColumnModel().getColumn(columnIndex:8).setCellEditor(new RatingCellEditor());
    addToLibraryTable.getColumnModel().getColumn(columnIndex:5).setCellEditor(new TimeCellEditor());
    addToLibraryTable.getColumnModel().getColumn(columnIndex:6).setCellEditor(new DateCellEditor(addToLibraryTable));
    addToLibraryTable.getColumnModel().getColumn(columnIndex:7).setCellEditor(new DateCellEditor(addToLibraryTable));
    JComboBox<String> statusComboBox = new JComboBox<>(new String[]{"Not started", "Ongoing", "Completed"});
    addToLibraryTable.getColumnModel().getColumn(columnIndex:4).setCellEditor(new DefaultCellEditor(statusComboBox));
```

```java
// Adding a button to delete rows
JButton deleteButton = new JButton(text:"Delete");
deleteButton.addActionListener(e -> {
    int row = addToLibraryTable.getSelectedRow();
    if (row != -1) {
        String title = (String) tableModel.getValueAt(row, column:0);
        deleteBookByUsernameAndTitle(username, title);
        tableModel.removeRow(row);
    } else {
        JOptionPane.showMessageDialog(parentComponent:null, message:"Please select a row to delete.", title:"No Row Selected", JOptionPane.WARNING_MESSAGE);
    }
});
addToLibraryTable.getColumnModel().getColumn(columnIndex:10).setCellRenderer(new ButtonRenderer());
addToLibraryTable.getColumnModel().getColumn(columnIndex:10).setCellEditor(new ButtonEditor(deleteButton));

// Scroll pane for table
JScrollPane scrollPane = new JScrollPane(addToLibraryTable);
getContentPane().add(scrollPane, BorderLayout.CENTER);

// Button to save the changes made to the table data
JButton saveButton = new JButton(text:"Save");
saveButton.addActionListener(e -> {
    if (isDataComplete()) {
        saveData(username);
    } else {
        JOptionPane.showMessageDialog(parentComponent:null, message:"Please complete all required fields.", title:"Incomplete Data", JOptionPane.WARNING_MESSAGE);
    }
});
saveButton.setBackground(new Color(r:34, g:139, b:34)); // Dark green
saveButton.setForeground(Color.WHITE);
getContentPane().add(saveButton, BorderLayout.SOUTH);
```

UI Snippets:

General Database for Admin:

Code Snippets:

```java
    /     Constructor to initialize the GUI components     /
    public GeneralDatabaseOfAdmin() {
        setTitle(title:"Admin Panel");
        setSize(width:800, height:600);
        setLocationRelativeTo(c:null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JTabbedPane tabbedPane = new JTabbedPane();
        tabbedPane.setBackground(Color.PINK);

        JPanel titlesPanel = new JPanel(new BorderLayout());
        titlesPanel.setBackground(Color.PINK);
        tabbedPane.addTab(title:"Books", titlesPanel);
        createTitlesPanel(titlesPanel);

        // Users tab
        JPanel usersPanel = new JPanel(new BorderLayout());
        usersPanel.setBackground(Color.PINK);
        tabbedPane.addTab(title:"Users", usersPanel);
        createUsersPanel(usersPanel);

        JPanel reviewsPanel = new JPanel(new BorderLayout());
        reviewsPanel.setBackground(Color.PINK);
        tabbedPane.addTab(title:"User Reviews", reviewsPanel);
        createUserReviewsPanel(reviewsPanel);

        add(tabbedPane);
        setVisible(b:true);
```

```java
// Header panel with BorderLayout
JPanel headerPanel = new JPanel(new BorderLayout());
headerPanel.setBackground(Color.PINK); // Set background color to pink

// Header label
JLabel headerLabel = new JLabel(text:"Book Management System");
headerLabel.setFont(new Font(name:"Arial", Font.BOLD, size:31));
headerLabel.setForeground(Color.BLACK);
headerLabel.setHorizontalAlignment(SwingConstants.CENTER);
headerPanel.add(headerLabel, BorderLayout.CENTER);

// Back button panel with FlowLayout
JPanel backButtonPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
backButtonPanel.setBackground(Color.PINK); // Set background color to pink

// Back button
JButton backButton = new JButton(text:"Back");
backButton.addActionListener(e -> {
    MyGUI loginPage = new MyGUI();
    loginPage.setVisible(b:true);
    dispose();
});
backButton.setBackground(new Color(r:0,g:128,b:255));
backButton.setForeground(Color.WHITE);
backButtonPanel.add(backButton);
```

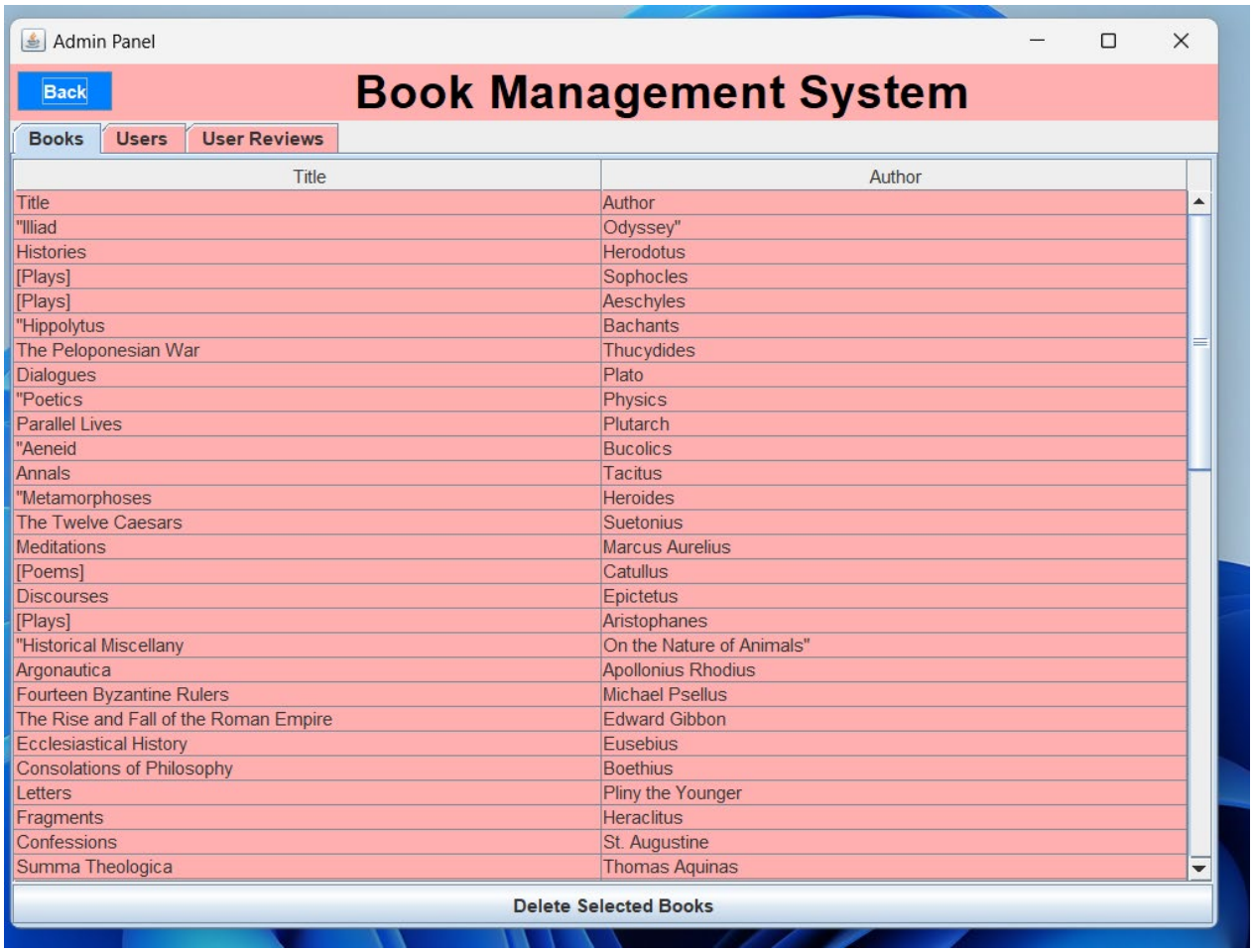
```java
// Add header panel and back button panel to NORTH of the frame
headerPanel.add(backButtonPanel, BorderLayout.WEST);
add(headerPanel, BorderLayout.NORTH);

// Search panel with FlowLayout
JPanel searchPanel = new JPanel(new FlowLayout(FlowLayout.RIGHT));
searchPanel.setBackground(Color.PINK); // Set background color to pink

// Add search panel to NORTH of the frame
headerPanel.add(searchPanel, BorderLayout.EAST);

// Initialize table model
tableModel = new DefaultTableModel() {
    @Override
    public Class<?> getColumnClass(int columnIndex) {
        if (columnIndex == 4) {
            return Boolean.class;
        } else {
            return String.class;
        }
    }

    @Override
    public boolean isCellEditable(int row, int column) {
        return column == 4;
    }
};
```

UI Snippets:

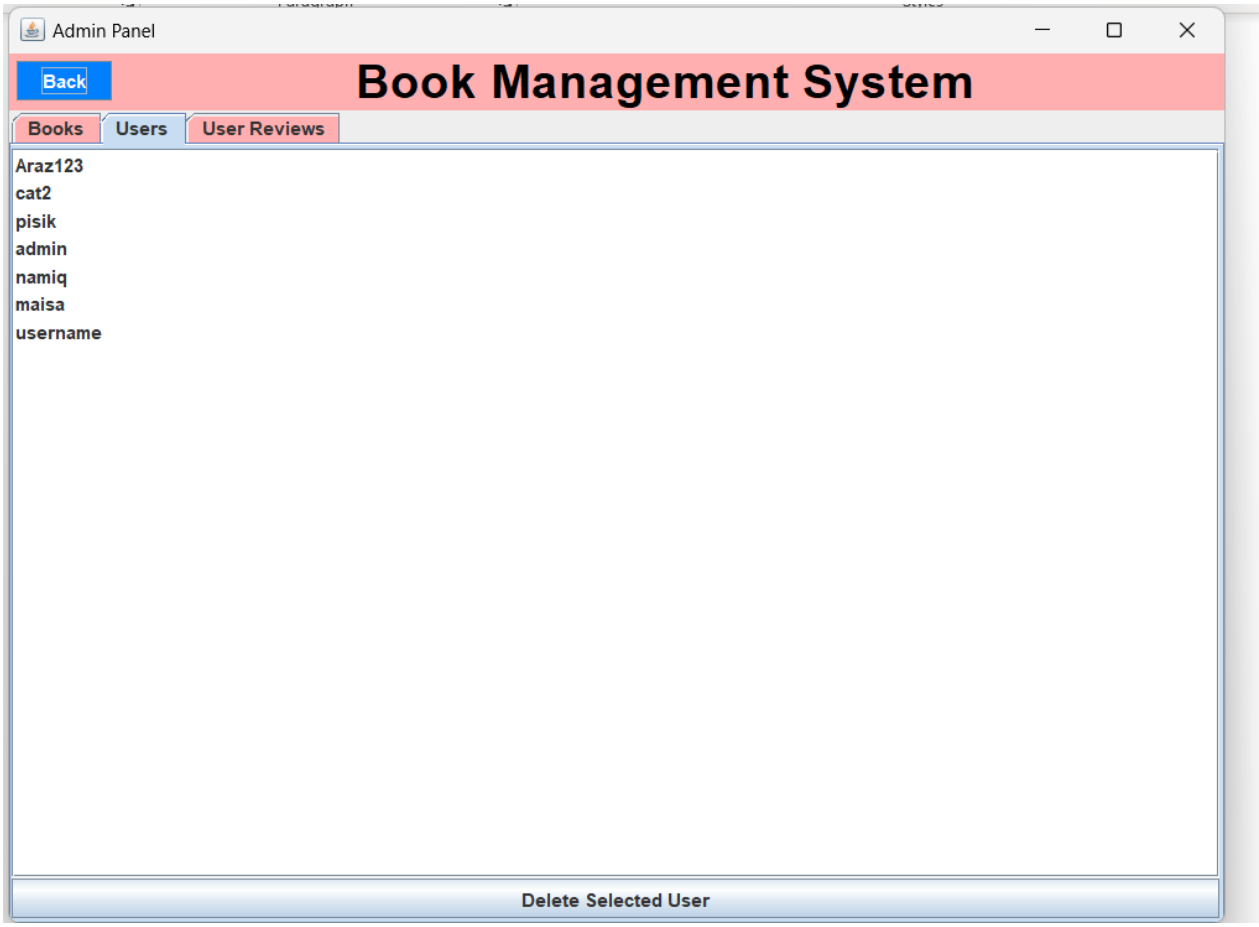
```java
/** Create panel for Users tab */
private void createUsersPanel(JPanel usersPanel) {
    // Create list to display users
    JList<String> userList = new JList<>();
    loadUsers(userList);
    JScrollPane scrollPane = new JScrollPane(userList);
    usersPanel.add(scrollPane, BorderLayout.CENTER);

    // Button to delete selected user
    JButton deleteButton = new JButton(text:"Delete Selected User");
    deleteButton.addActionListener(e -> deleteUser(userList.getSelectedValue()));
    usersPanel.add(deleteButton, BorderLayout.SOUTH);
}

/** Load users into the JList */
private void loadUsers(JList<String> userList) {
    allUsers = new ArrayList<>();
    DefaultListModel<String> model = new DefaultListModel<>();
    // Load users from CSV file
    try (BufferedReader br = new BufferedReader(new FileReader(fileName:"users.csv"))) {
        String line;
        while ((line = br.readLine()) != null) {
            String[] data = line.split(regex:",");
            allUsers.add(data);
            model.addElement(data[0]); // Add username to the list
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
    userList.setModel(model);
}
```

```java
/** Create panel for Titles tab */
private void createTitlesPanel(JPanel titlesPanel) {
    titlesTableModel = new DefaultTableModel(new Object[]{"Title", "Author"}, rowCount:0);
    JTable table = new JTable(titlesTableModel);
    table.setDefaultRenderer(columnClass:Object.class, new CustomRenderer());

    JScrollPane scrollPane = new JScrollPane(table);
    titlesPanel.add(scrollPane, BorderLayout.CENTER);

    loadTitles();
    JButton deleteButton = new JButton(text:"Delete Selected Books");
    deleteButton.addActionListener(e -> deleteSelectedTitles(table.getSelectedRows()));
    titlesPanel.add(deleteButton, BorderLayout.SOUTH);
}

/** Method to delete selected titles */
private void deleteSelectedTitles(int[] selectedRows) {
    for (int i = selectedRows.length - 1; i >= 0; i--) {
        int selectedRow = selectedRows[i];
        titlesTableModel.removeRow(selectedRow);
        allBooks.remove(selectedRow); // Remove corresponding data from the list
    }
    saveTitlesToCSV(); // Update CSV file with modified data
}
```

UI Snippets:



Title	Author
Title	Author
"Illiad	Odyssey"
Histories	Herodotus
[Plays]	Sophocles
[Plays]	Aeschyles
"Hippolytus	Bachants
The Peloponesian War	Thucydides
Dialogues	Plato
"Poetics	Physics
Parallel Lives	Plutarch
"Aeneid	Bucolics
Annals	Tacitus
"Metamorphoses	Heroides
The Twelve Caesars	Suetonius
Meditations	Marcus Aurelius
[Poems]	Catullus
Discourses	Epictetus
[Plays]	Aristophanes
"Historical Miscellany	On the Nature of Animals"
Argonautica	Apollonius Rhodius
Fourteen Byzantine Rulers	Michael Psellus
The Rise and Fall of the Roman Empire	Edward Gibbon
Ecclesiastical History	Eusebius
Consolations of Philosophy	Boethius
Letters	Pliny the Younger
Fragments	Heraclitus
Confessions	St. Augustine
Summa Theologica	Thomas Aquinas


Admin Panel
Back
Book Management System
Books
Users
User Reviews
Araz123
cat2
pisik
admin
namiq
maisa
username
Delete Selected User

Admin Panel

Back

Book Management System

Minimize

Books | Users | User Reviews

Title	Author	Rating	Reviews
Title	Author	No Rating	No reviews
Illiad	Unknown	No Rating	No reviews
Odyssey	Unknown	No Rating	No reviews
Histories	Herodotus	No Rating	No reviews
[Plays]	Sophocles	No Rating	No reviews
[Plays]	Aeschyles	No Rating	No reviews
"Hippolytus	Bachants	No Rating	cat2
The Peloponesian War	Thucydides	No Rating	No reviews
Dialogues	Plato	No Rating	pisik
"Poetics	Physics	No Rating	No reviews
Parallel Lives	Plutarch	No Rating	No reviews
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The Twelve Caesars	Suetonius	No Rating	No reviews
Meditations	Marcus Aurelius	No Rating	No reviews
[Poems]	Catullus	No Rating	No reviews
Discourses	Epictetus	No Rating	No reviews
[Plays]	Aristophanes	No Rating	No reviews
Historical Miscellany	Unknown	No Rating	No reviews
On the Nature of Animals	Unknown	No Rating	No reviews
Argonautica	Apollonius Rhodius	No Rating	No reviews
Fourteen Byzantine Rulers	Michael Psellus	No Rating	No reviews
The Rise and Fall of the Roman Em	Edward Gibbon	No Rating	No reviews
Ecclesiastical History	Eusebius	No Rating	No reviews
Consolations of Philosophy	Boethius	No Rating	No reviews
Letters	Pliny the Younger	No Rating	No reviews
Fragments	Heraclitus	No Rating	No reviews

Delete Selected Reviews

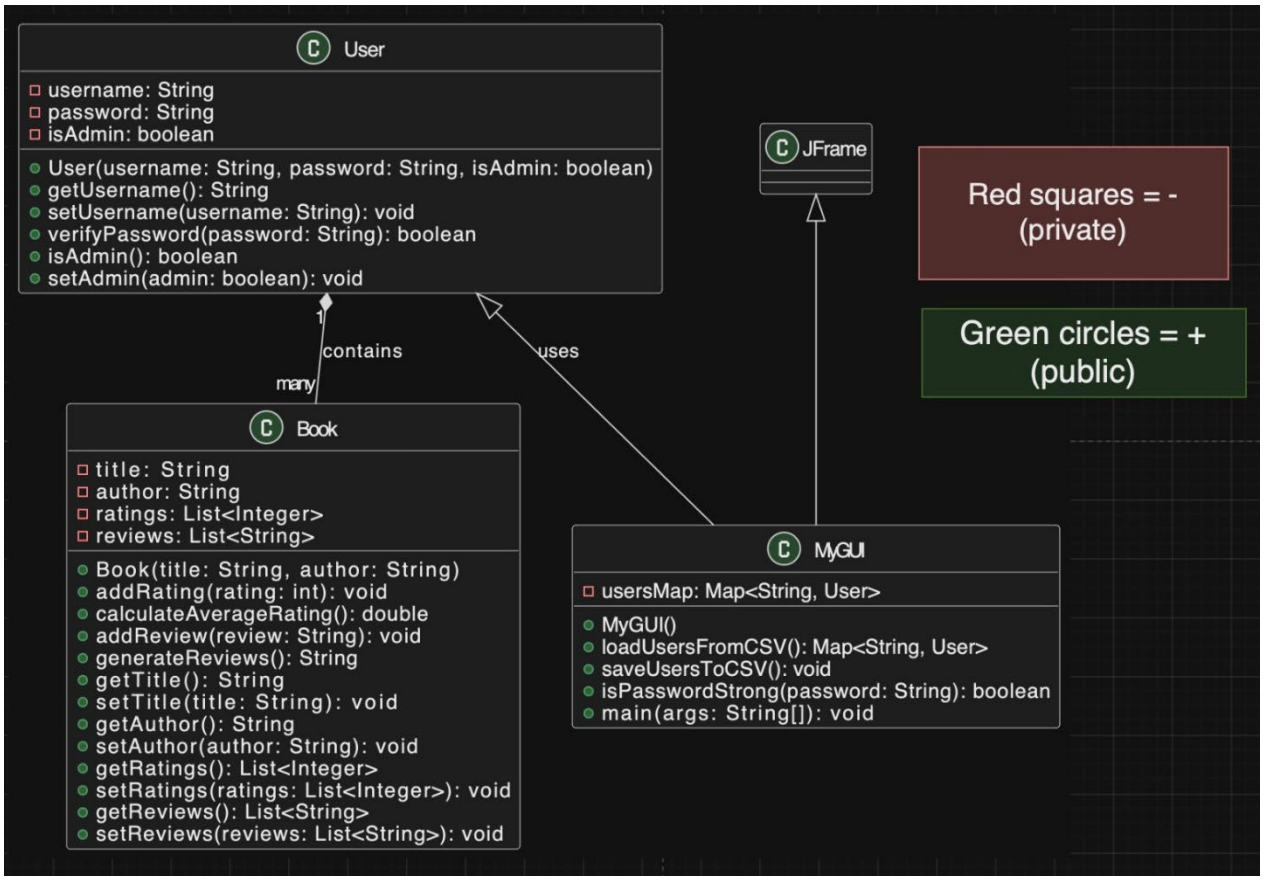
Work Allocation:

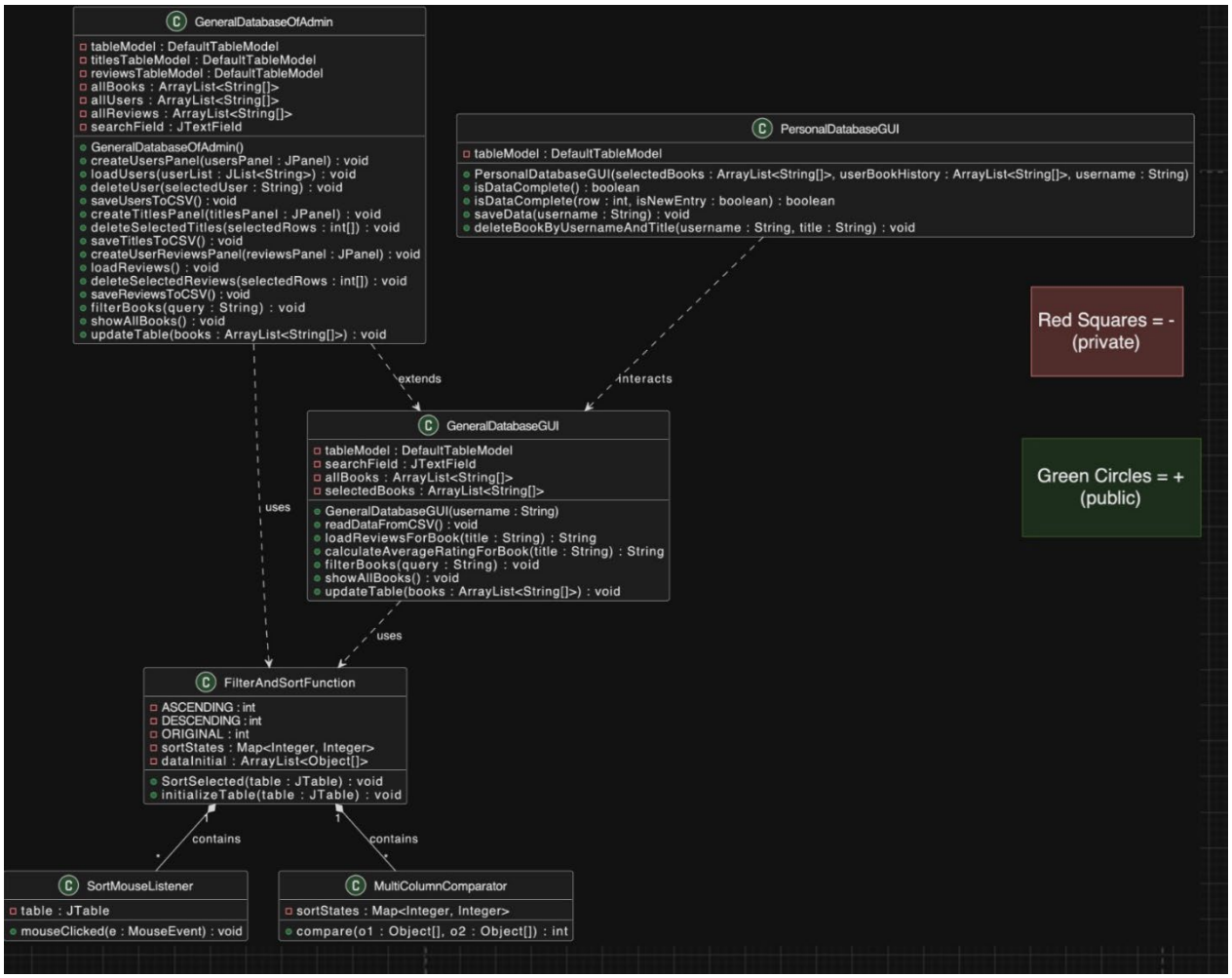
Team Members:	Contribution:	Responsibilities:
Maisa Babayeva	32.5%	Did login/registration process, solved user/admin issues, solved problems related to personal and general database, wrote README file, wrote report, worked with CRUD operations
Araz Valishov	22.5%	Prepared UML diagrams, worked with personal database, wrote report
Nahid Aghayev	22.5%	Worked with general database, worked with filter and sorting, wrote report
Javid Eynullayev	22.5%	Worked with user/admin part, wrote report

Video Link: https://youtu.be/xkWsiIDcBbw?feature=shared

Github Link: **https://github.com/ADA-SITE-CSCI-1202/team-project-team-28**

UML Diagrams:


User
username: String
password: String
isAdmin: boolean
User(username: String, password: String, isAdmin: boolean)
getUsername(): String
setUsername(username: String): void
verifyPassword(password: String): boolean
isAdmin(): boolean
setAdmin(admin: boolean): void
JFrame
Red squares = - (private)
Green circles = + (public)
1
contains
many
uses
Book
title: String
author: String
ratings: List<Integer>
reviews: List<String>
Book(title: String, author: String)
addRating(rating: int): void
calculateAverageRating(): double
addReview(review: String): void
generateReviews(): String
getTitle(): String
setTitle(title: String): void
getAuthor(): String
setAuthor(author: String): void
getRatings(): List<Integer>
setRatings(ratings: List<Integer>): void
getReviews(): List<String>
setReviews(reviews: List<String>): void
MyGUI
usersMap: Map<String, User>
MyGUI()
loadUsersFromCSV(): Map<String, User>
saveUsersToCSV(): void
isPasswordStrong(password: String): boolean
main(args: String[]): void



Red squares show private access modifier, Green squares show public access modifier

Note: There is also one more file in our GitHub account which is not used: TransitionPage.java. This file was used before current design and was opened after login/registration page. However, as a team,we thought that it is better to open directly GeneralDatabaseGUI file.